CSD 340 practice problems for LMM - HW5

1. You observed a single sample Z which follows the Geometric(p) distribution, with unknown $p \in [0, 1]$. Consider a likelihood Ratio test with $H_0: p \leq c_0$ for some $p_0 \in [0, 1]$ versus $H_1: p > c_0$.

   a) Find the likelihood ratio statistics.
   b) For any $\alpha \in (0, 1)$, find the rejection region of the test with size at most α.

2. You observed a sample of $Z_1, Z_2, \ldots, Z_n$ from Poisson distribution. Denote the unknown parameter of the Poisson distribution as λ, and you compare two possible values (c_0 and c_1) for λ. Perform a likelihood ratio test with the Null hypothesis $\lambda = c_0$ versus the alternative hypothesis $\lambda = c_1$, where $c_0 < c_1$.

   a) For any $\alpha \in (0, 1)$, find the rejection region of the test with size at most α.
   b) Let $c_0 = 2$ and $c_1 = 3$ and a sample of size 5, find the test with size at most 0.05.

3. Given the RSS of the following models after fitting a sample of size 24, answer the following questions.

Model	RSS
M1: $lm(y \sim 1)$	184066
M2: $lm(y \sim X_1)$	181544.5
M3: $lm(y \sim X_2)$	155638
M4: $lm(y \sim X_1 + X_2)$	87246.89

   a) Complete the Analysis of Variance table of M4:

	df	Sum of squares	F value	Pr(>F)
X_1				
X_2				
Residuals			NA	NA

b) Complete the output table of R command drop1(M4):

	df	Sum of Squares	RSS
<none>			
X_1			
X_2			

c) Perform the F-test with Null hypothesis $\beta_1 = \beta_2 = 0$ versus the opposite.

4. Consider the #7 in HW4. From the data analysis performed in HW4, design a LMM for the data. You should consider the variation of MathAch mean among different subgroups observed. Specify your model by HLM as well as corresponding LMM.

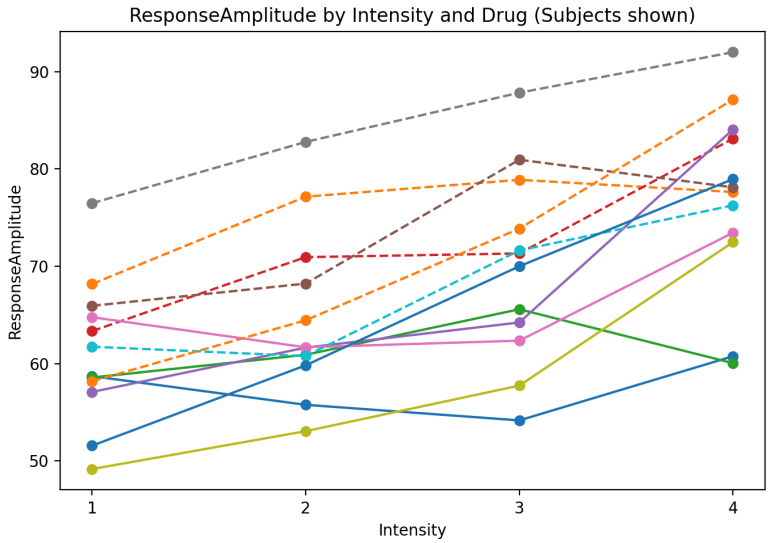
5. A lab study measures ResponseAmplitude for 6 subjects under:
- Drug ∈ {Control, Drug}
- Intensity ∈ {1,2,3,4}

Each subject is measured at every Drug × Intensity combination. The response variable is ResponseAmplitude.

Some evidence is given below.

Evidence A: Raw Profile Plot

In the plot, Colored lines represent different subjects. Solid = Control, Dashed = Drug.



Evidence B: Cell Means

> with(dat, tapply(ResponseAmplitude, list(Drug, Intensity), mean))

Intensity	1	2	3	4
Control	55.80	58.54	63.87	67.44
Drug	63.94	73.51	75.15	82.99

Evidence C: SD by Intensity (pooled)

> aggregate(ResponseAmplitude ~ Intensity, data = dat, sd)

Intensity	ResponseAmplitude
1	7.1312
2	9.1598
3	11.5567
4	12.9539

Evidence D: Subject-level Drug Differences

> dwide[, c("Subject","DrugDiff")]

Subject	DrugDiff
S1	11.47
S2	11.30
S3	18.49

```
S4      16.83
S5      5.06
S6      13.25
```

Evidence E: Subject-specific Intensity Slopes (SD)

(For each subject and each Drug level: lm(ResponseAmplitude ~ Intensity) and report the slope.)

```
> slope_tab

  Subject    Drug  Slope

1       S1 Control   4.95

2       S1    Drug    5.10

3       S2 Control   4.88

4       S2    Drug    5.02

5       S3 Control   5.06

6       S3    Drug    4.97

7       S4 Control   4.92

8       S4    Drug    5.08

9       S5 Control   5.01

10      S5    Drug    4.93

11      S6 Control   4.97

12      S6    Drug    5.04


> with(slope_tab, tapply(Slope, Drug, sd))

  Control      Drug

0.0671     0.0679
```

Evidence F: part of a tested Model output

The following quantities were estimated from a model allowing:
- correlation across Intensity within Subject,
- different residual variances by Intensity level.

Correlation parameter:
Phi = 0.62
Variance multipliers by Intensity:

1	2	3	4
1.00	1.07	1.46	1.92

Residual standard error (baseline level):
3.20

Based on the evidence above, answer the questions below and reference specific evidence)

(a) Specify the fixed-effects structure and justify:

- Include or exclude: Drug, Intensity, Drug×Intensity.
- Decide whether Intensity should be treated as numeric or factor; justify briefly.

(b) Specify the random-effects structure by deciding whether to include each of the following subject-level random effects and justifying your choice:

- random intercept
- random slope for Drug
- random slope for Intensity

(c) Specify the residual covariance structure.

(d) Write your final LMM in R formula syntax. (You may use nlme-style notation.)

(e) Covariance matrices G and R. For your final model, provide both:

- Theoretical format: write G and R in symbolic (parameterized) matrix form.
- Detailed numeric matrix: using Evidence F, compute the numeric 4×4 matrix R for Intensity = 1,2,3,4 within one subject and one Drug condition.